Simultaneous Equations and Inequalities

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Simultaneous Equations & Inequalities Practice Exercises**

[Simultaneous Equations & Inequalities · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

2 **[Foundational]** • **Target: Simultaneous Equations & Inequalities Practice Exercises**

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

Find the critical values where $2x^2 - 7x + 3 = 0$. Factor the quadratic: $(2x - 1)(x - 3) = 0$, so the roots are $x = \frac{1}{2}$ and $x = 3$. Since the coefficient of x^2 is positive ($a = 2 > 0$), the parabola opens upward. The quadratic is less than or equal to zero between and including the roots. Therefore, the solution set is $\frac{1}{2} \leq x \leq 3$.

3 **[Foundational]** • **Target: Simultaneous Equations & Inequalities Practice Exercises**

[Differential Equations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

4 **[Foundational]** • **Target: Simultaneous Equations & Inequalities Practice Exercises**

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

Separate variables and integrate: $\int \frac{1}{\theta - 20}\, d\theta = -\int k\, dt \Rightarrow \ln(\theta - 20) = -kt + C$ Apply initial condition $t = 0, \theta = 80$: $\ln(60) = C \Rightarrow \ln\left(\frac{\theta - 20}{60}\right) = -kt \Rightarrow \theta - 20 = 60e^{-kt}$ Apply condition $t = 10, \theta = 50$: $30 = 60e^{-10k} \Rightarrow e^{-10k} = \frac{1}{2} \Rightarrow k = \frac{\ln 2}{10} \approx 0.06931$ Set $\theta = 30$.

5 **[Foundational]** • **Target: Simultaneous Equations & Inequalities Practice Exercises**

[Differentiation & Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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6 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

[Trigonometry · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

7 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

[About AS Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

[Binomial Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

[Coordinate Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

[Representation of Data · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

[Discrete Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

[Energy, Work & Power · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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13 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

[Forces & Equilibrium · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

14 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

[Graphs & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

[Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

16 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

[Newton's Laws of Motion · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

[Normal Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

[Circular Measure · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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19 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

[Series & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

[Statistics & Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.

a. (1, 2)

b. **(2, 1)** ✓

c. (3, 0)

d. (0, 3)

The point of intersection of two lines is the solution to the system of simultaneous equations that represent these lines. By visually inspecting the graph, we can identify the coordinates where the two lines cross. The first line appears to pass through the points (0, 3) and (3, 0). The second line appears to pass through the points (0, 1) and (2, 3). Let's find the equations of the lines. For the first line: Gradient $m_1 = \frac{0-3}{3-0} = \frac{-3}{3} = -1$. Using the point-slope form with (3, 0).

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22 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

a. c = 3, k = 4 ✓

b. c = 5, k = 3

c. c = 3, k = 3

d. c = 5, k = 4

The points of intersection satisfy both equations. Therefore, we can set the equations equal to each other: $2x + c = -x^2 + 4x + 5$ Rearranging this equation gives: $x^2 - 2x + (c - 5) = 0$ We are given that the graphs intersect at $x = -1$. Substituting $x = -1$ into the quadratic equation $y = -x^2 + 4x + 5$: $y = -(-1)^2 + 4(-1) + 5 = -1 - 4 + 5 = 0$ So, the point of intersection is $(-1, 0)$. Since this point also lies on the line $y = 2x + c$, we can substitute $x = -1$ and $y = 0$ into the linear equation.

23 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Find the x-coordinates of the points of intersection of $y = x^2 - 9x + 22$ and $y = 2$.

a. x=4 or x=5 ✓

b. x=-4 or x=-5

c. x=5 or x=5

d. x=4 or x=4

Substituting the line into the curve gives $x^2 - 9x + 20 = 0$, which factorises as $(x - 5)(x - 4) = 0$, so $x = 4$ or $x = 5$.

24 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the simultaneous equations $2x + 3y = 18$ and $3x - 4y = 10$.

a. x=7, y=2

b. x=-6, y=-2

c. x=6, y=2 ✓

d. x=2, y=6

Solving simultaneously (elimination or substitution) gives $x = 6$, $y = 2$.

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25 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

A curve has equation $y = x\sqrt{16 - x^2}$ for $0 \le x \le 4$. A point $P(x, y)$ moves along the curve in the first quadrant, and the normal to the curve at P intersects the x-axis at the point N. Given that the x-coordinate of P is increasing at a constant rate of $\sqrt{3}$ units s^{-1}, find the rate of change of the area of triangle OPN, where O is the origin, at the instant when $x = 2$.

a. -6 units^2 s^{-1} ✓

b. 6 units^2 s^{-1}

c. -12 units^2 s^{-1}

d. $-2\sqrt{3}$ units^2 s^{-1}

Step 1: Differentiate $y = x\sqrt{16 - x^2}$ using the product rule: $\frac{dy}{dx} = 1 \cdot \sqrt{16 - x^2} + x \cdot \frac{-2x}{2\sqrt{16 - x^2}} = \frac{16 - 2x^2}{\sqrt{16 - x^2}}$ At $x = 2$: $y = 2\sqrt{16 - 4} = 4\sqrt{3}$ $\frac{dy}{dx} = \frac{16 - 2(4)}{\sqrt{12}} = \frac{8}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$ Step 2: Find the coordinates of N, the x-intercept of the normal. The gradient of the normal at P is: $m_N = -\frac{1}{\frac{dy}{dx}} = -\frac{\sqrt{16 - x^2}}{16 - 2x^2}$ The equation of the no.

26 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

A sector OAB of a circle with centre O and radius 12 cm has angle $\angle AOB = \frac{\pi}{2}$ radians. The point C is the midpoint of the radius OA. The line passing through C perpendicular to OA intersects the circular arc AB at the point D. The straight line segment BC is drawn. Find the exact area, in cm^2, of the region bounded by the line segment CD, the circular arc BD, and the line segment BC.

a. $12\pi + 18\sqrt{3} - 36$ ✓

b. $24\pi + 18\sqrt{3} - 36$

c. $12\pi + 36\sqrt{3} - 36$

d. $18\pi + 18\sqrt{3} - 36$

1. **Analyze the geometry of right triangle OCD**: - C is the midpoint of OA, so $OC = \frac{1}{2}(12) = 6$ cm. - D lies on the arc of the circle, so the radius $OD = 12$ cm. - Since $CD \perp OA$, triangle OCD is a right-angled triangle at C. - $\cos(\angle COD) = \frac{OC}{OD} = \frac{6}{12} = \frac{1}{2} \Rightarrow \angle COD = \frac{\pi}{3}$ radians. 2. **Find the side length CD and the area of ΔOCD**.

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27 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

In the expansion of $(1+ax)^n$ in ascending powers of x, the coefficient of x^2 is 60 and the coefficient of x^3 is 160, where n is a positive integer and $a>0$. A cubic polynomial $P(x)$, in which the coefficient of x^3 is 1, has $(x-a)^2$ as a factor and leaves a remainder of n when divided by $(x-1)$. Find the remainder when $P(x)$ is divided by $(x+a)$.

a. **48 ✓**

b. -48

c. -112

d. 0

Step 1: Determine n and a using the Binomial Theorem. In the expansion of $(1+ax)^n$: Term in $x^2=\binom{n}{2}(ax)^2=\frac{n(n-1)}{2}a^2x^2=60x^2\Rightarrow n(n-1)a^2=120$ Term in $x^3=\binom{n}{3}(ax)^3=\frac{n(n-1)(n-2)}{6}a^3x^3=160x^3\Rightarrow n(n-1)(n-2)a^3=960$ Dividing the equation for x^3 by the equation for x^2: $\frac{n(n-1)(n-2)a^3}{n(n-1)a^2}=\frac{960}{120}\Rightarrow(n-2)a=8\Rightarrow a=\frac{8}{n-2}$ Substi.

28 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

The line $y=7x-4$ is tangent to the curve $y=ax^2+\frac{b}{x}$ at the point where $x=2$, where a and b are constants and $xe0$. Find the coordinates of the stationary point of the curve and determine its nature.

a. $(1,6)$**, which is a local minimum ✓**

b. $(1,6)$, which is a local maximum

c. $(-1,\ -2)$, which is a local minimum

d. $(2,10)$, which is a local minimum

1. **Find the point of tangency:** Substitute $x=2$ into the tangent line equation $y=7x-4$: $y=7(2)-4=10$ Since the line touches the curve at $x=2$, the point $(2,10)$ lies on the curve $y=ax^2+\frac{b}{x}$: $a(2)^2+\frac{b}{2}=10\Rightarrow4a+\frac{b}{2}=10\Rightarrow8a+b=20\quad---\ (1)$ 2. **Use the gradient of the tangent:** Differentiating $y=ax^2+bx^{-1}$ gives: $\frac{dy}{dx}=2ax-\frac{b}{x^2}$ The gradient of the tangent line $y=7x-4$ is 7. At.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.

a. **108** ✓

b. 84

c. 64

d. 12

To find the area of the enclosed region, we synthesize differential calculus, polynomial algebra, and definite integration: 1. **Find the equation of the tangent line at $P(2, 4)$:** Differentiating the curve equation: $\frac{dy}{dx} = 3x^2 - 3$ At $x = 2$, the gradient of the tangent is: $m = 3(2)^2 - 3 = 12 - 3 = 9$ Using the point-slope formula with $P(2, 4)$: $y - 4 = 9(x - 2) \Rightarrow y = 9x - 14$ 2. **Find the intersection point Q:** Set the curve and tangent equations equal.

30 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2. During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm.

a. 20π ✓

b. 50π

c. $\frac{36\pi}{5}$

d. 11π

1. **Express the surface area constraint:** The total exterior surface area consists of the flat circular base, the curved cylindrical wall, and the hemispherical dome: $S = \pi r^2 + 2\pi rh + 2\pi r^2 = 3\pi r^2 + 2\pi rh$ Given $S = 180\pi$: $3\pi r^2 + 2\pi rh = 180\pi \Rightarrow 3r^2 + 2rh = 180$ 2. **Find the value of r when $h = 16.5$ cm:** $3r^2 + 2(16.5)r = 180 \Rightarrow 3r^2 + 33r - 180 = 0$ Dividing by 3: $r^2 + 11r - 60 = 0 \Rightarrow (r + 15)(r - 4) = 0$ Since $r > 0$, we have.

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31 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

The variables x and y, where $x > 0$, satisfy the relationship $y = ax + \frac{b}{x}$ where a and b are positive constants. When xy is plotted on the vertical axis against x^2 on the horizontal axis, a straight line is obtained which passes through the points $(2, 11)$ and $(5, 23)$. Given further that x and y satisfy the logarithmic equation $\log_3(y - ax) + \log_9(x^2y^2) = 1 + \log_3(x + 10)$ find the sum of all possible values of x.

a. $\frac{10}{3}$ ✓

b. $\frac{8}{3}$

c. $\frac{7}{2}$

d. $\frac{13}{3}$

We proceed in three main stages: **Stage 1: Straight-Line Reduction to determine a and b** Multiplying $y = ax + \frac{b}{x}$ by x yields: $xy = a(x^2) + b$ Letting $Y = xy$ and $X = x^2$, the equation is in the linear form $Y = aX + b$. The line passes through $(2, 11)$ and $(5, 23)$. The gradient is: $a = \frac{23-11}{5-2} = \frac{12}{3} = 4$ Substituting into $Y = 4X + b$ using $(2, 11)$: $11 = 4(2) + b \Rightarrow b = 3$ Thus, $y = 4x + \frac{3}{x}$, which implies $y - 4x = \frac{3}{x}$ and.

32 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

A group of n students, which includes Ali and Priya, are to be arranged in a straight line for a photograph. The number of different arrangements in which Ali and Priya are not standing next to each other is 3600. Find the value of the integer n.

a. 6

b. **7** ✓

c. 8

d. 9

We can find the number of arrangements where Ali and Priya do not stand next to each other using the complementary method or the insertion (gap) method. **Method 1: Complementary Counting** 1. The total number of unrestricted arrangements of the n students in a straight line is $n!$. 2. Consider the arrangements in which Ali and Priya *do* stand next to each other: - Treat Ali and Priya as a single block. There are now $(n - 1)$ entities to arrange, which can be done in $(n - 1)!$ ways. - Within the block, Ali an.

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33 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

Relative to a fixed origin O, ship A has an initial position vector of $(24\mathbf{i} + 10\mathbf{j})$ km and moves with a constant velocity of $(-2\mathbf{i} + 5\mathbf{j})$ km h^{-1}. Ship B has an initial position vector of $(-20\mathbf{i} - 18\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 12\mathbf{j})$ km h^{-1}, where u is a constant. Given that ship B intercepts ship A, find the speed of ship B.

a. 15 km h^{-1} ✓

b. 9 km h^{-1}

c. 21 km h^{-1}

d. 34 km h^{-1}

Let t be the time in hours after the ships begin moving. The position vector of ship A at time t is: $\mathbf{r}_A(t) = (24\mathbf{i} + 10\mathbf{j}) + t(-2\mathbf{i} + 5\mathbf{j}) = (24 - 2t)\mathbf{i} + (10 + 5t)\mathbf{j}$ The position vector of ship B at time t is: $\mathbf{r}_B(t) = (-20\mathbf{i} - 18\mathbf{j}) + t(u\mathbf{i} + 12\mathbf{j}) = (-20 + ut)\mathbf{i} + (-18 + 12t)\mathbf{j}$ For ship B to intercept ship A, their positions must be identical at some time $t > 0$, so.

34 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

Two parallel lines, L_1 and L_2, have n and $n + 2$ distinct points marked on them respectively, where n is an integer with $n \geq 2$. Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear. Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is $6 : 5$, find the total number of triangles that can be formed.

a. 175 ✓

b. 210

c. 220

d. 96

1. **Determine the number of triangles, T:** Three points form a triangle if and only if they are not all on the same straight line. - Case 1: 1 point from L_1 and 2 points from L_2: $\binom{n}{1}\binom{n+2}{2} = n \cdot \frac{(n+2)(n+1)}{2}$ - Case 2: 2 points from L_1 and 1 point from L_2: $\binom{n}{2}\binom{n+2}{1} = \frac{n(n-1)}{2} \cdot (n+2)$ Summing both cases: $T = \frac{n(n+2)}{2}[(n+1) + (n-1)] = \frac{n(n+2)}{2}(2n) = n^2(n+2)$ 2. **Determine the number of quadrilaterals, Q.

35 **[Intermediate]** • **Target: Simultaneous Equations & Inequalities Practice Exercises**

In triangle ABC, the side lengths are $AB = (x + 2)$ cm, $BC = (3x - 1)$ cm, and $AC = 7$ cm. Given that $\angle ABC = 60^\circ$ and $x > 0$, find the exact area of triangle ABC.

a. $10\sqrt{3}$ cm² ✓

b. $5\sqrt{3}$ cm²

c. $20\sqrt{3}$ cm²

d. 10 cm²

Step 1: Use the cosine rule on triangle ABC to set up an equation for x:
$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(\angle ABC)$ Substitute the given values and $\cos(60^\circ) = \frac{1}{2}$:
$7^2 = (x + 2)^2 + (3x - 1)^2 - 2(x + 2)(3x - 1)\left(\frac{1}{2}\right)$ Step 2: Expand each term: $(x + 2)^2 = x^2 + 4x + 4$
$(3x - 1)^2 = 9x^2 - 6x + 1$ $(x + 2)(3x - 1) = 3x^2 + 5x - 2$ Substitute these back into the equation:
$49 = (x^2 + 4x + 4) + (9x^2 - 6x + 1) - (3x^2 + 5x - 2)$ $49 = 10x^2 - 2x + 5 - 3x^2 - 5x + 2$.

36 **[Intermediate]** • **Target: Simultaneous Equations & Inequalities Practice Exercises**

Variables x and y satisfy the equation $y = Ax^k$, where A and k are constants, with $x > 0$. The graph of $\ln y$ against $\ln x$ is a straight line passing through the points $(\ln 2, \ln 72)$ and $(\ln 6, \ln 8)$. Find the value of x when $y = 18$.

a. **4** ✓

b. 2

c. 1

d. 8

Step 1: Reduce the non-linear relationship to linear form. Taking the natural logarithm of both sides of $y = Ax^k$: $\ln y = \ln(Ax^k) = \ln A + k\ln x$ This is in the linear form $Y = mX + c$, where $Y = \ln y$, $X = \ln x$, gradient $m = k$, and vertical intercept $c = \ln A$. Step 2: Determine the gradient k. Using the given points $(\ln 2, \ln 72)$ and $(\ln 6, \ln 8)$.

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37 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

The variables x and y are connected by the equation $y = \sqrt{2x+5}$. Given that x increases at a constant rate of 6 units per second, find the rate of change of y with respect to time when $x = 2$.

a. 2 units per second ✓

b. 1 unit per second

c. $\frac{1}{3}$ units per second

d. 12 units per second

We are given $y = (2x+5)^{\frac{1}{2}}$ and $\frac{dx}{dt} = 6$. Step 1: Differentiate y with respect to x using the chain rule: $\frac{dy}{dx} = \frac{1}{2}(2x+5)^{-\frac{1}{2}} \cdot \frac{d}{dx}(2x+5) = \frac{1}{2\sqrt{2x+5}} \cdot 2 = \frac{1}{\sqrt{2x+5}}$ Step 2: Evaluate $\frac{dy}{dx}$ at $x = 2$: $\frac{dy}{dx}\Big|_{x=2} = \frac{1}{\sqrt{2(2)+5}} = \frac{1}{\sqrt{9}} = \frac{1}{3}$ Step 3: Use.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

The variables x and y satisfy the relationship $y = Ax^n$, where A and n are constants. When $\log_2 y$ is plotted against $\log_2 x$, a straight line is obtained which passes through the points $(1, 5)$ and $(4, -1)$. What is the value of y when $x = 4$?

a. 8 ✓

b. $\frac{1}{2}$

c. 3

d. 32

1. Express the non-linear relationship in linear form $Y = mX + c$: Taking base-2 logarithms of both sides of $y = Ax^n$: $\log_2 y = \log_2(Ax^n) = n\log_2 x + \log_2 A$ Letting $Y = \log_2 y$ and $X = \log_2 x$, the equation becomes: $Y = nX + \log_2 A$ where the gradient is $m = n$ and the vertical intercept is $c = \log_2 A$. 2. Find the equation of the straight line: Using the given points $(X_1, Y_1) = (1, 5)$ and $(X_2, Y_2) = (4, -1)$: $m = \frac{-1-5}{4-1} = \frac{-6}{3} = -2$ Using the point.

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39 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $x - 2$ and leaves a remainder of -24 when divided by $x + 1$, where a and b are constants. Find the remainder when $P(x)$ is divided by $x - 3$.

a. 12 ✓

b. -30

c. 60

d. -150

Step 1: Use the Factor Theorem for the factor $x - 2$. Since $x - 2$ is a factor, $P(2) = 0$: $P(2) = 2(2)^3 + a(2)^2 + b(2) - 6 = 0$ $16 + 4a + 2b - 6 = 0 \Rightarrow 4a + 2b = -10 \Rightarrow 2a + b = -5$ $---$ (1) Step 2: Use the Remainder Theorem for division by $x + 1$. Since the remainder is -24, $P(-1) = -24$: $P(-1) = 2(-1)^3 + a(-1)^2 + b(-1) - 6 = -24$ $-2 + a - b - 6 = -24 \Rightarrow a - b - 8 = -24 \Rightarrow a - b = -16$ $---$ (2) Step 3: Solve the simultaneous equations.

40 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 18$. The particle is at instantaneous rest when $t = 1$. Given that the displacement of the particle from a fixed point O is $s = -10$ m when $t = 2$, find the displacement of the particle, in metres, when it is next at instantaneous rest.

a. -37 ✓

b. -25

c. -13

d. -5

1. **Find the velocity function $v(t)$:** $v(t) = \int a\,dt = \int (6t - 18)\,dt = 3t^2 - 18t + c$ Since the particle is at instantaneous rest when $t = 1$, $v(1) = 0$: $3(1)^2 - 18(1) + c = 0 \Rightarrow -15 + c = 0 \Rightarrow c = 15$ Thus, $v(t) = 3t^2 - 18t + 15$. 2. **Find when the particle is next at instantaneous rest:** Set $v(t) = 0$: $3(t^2 - 6t + 5) = 0 \Rightarrow 3(t - 1)(t - 5) = 0$ So the particle is at rest at $t = 1$ and next at $t = 5$. 3. **Find the displacement function $s(t)$:**.

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41 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

a. **13** ✓

b. 10

c. 8

d. 7

To form a committee of 5 members: 1. **Committees with exactly 3 men:** Since there are 5 members in total, the committee must contain 3 men and $5 - 3 = 2$ women.
Number of ways $= \binom{n}{3} \times \binom{6}{2} = \binom{n}{3} \times \frac{6 \times 5}{2} = 15\binom{n}{3}$ 2. **Committees with exactly 4 men:** The committee must contain 4 men and $5 - 4 = 1$ woman. Number of ways $= \binom{n}{4} \times \binom{6}{1} = 6\binom{n}{4}$ 3. **Equating the two expressions:**.

42 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

In the expansion of $(1 + ax)^n$ in ascending powers of x, where n is a positive integer and a is a constant, the coefficient of x is -12 and the coefficient of x^2 is 60. Find the coefficient of x^3.

a. -160 ✓

b. 160

c. -40

d. -240

Using the binomial theorem, the expansion of $(1 + ax)^n$ is given by:
$(1 + ax)^n = 1 + \binom{n}{1}(ax) + \binom{n}{2}(ax)^2 + \binom{n}{3}(ax)^3 + \ldots$ 1. **Equate the coefficient of x:**
$\binom{n}{1}a = na = -12 \Rightarrow a = -\frac{12}{n}$ 2. **Equate the coefficient of x^2:**
$\binom{n}{2}a^2 = \frac{n(n-1)}{2}a^2 = 60 \Rightarrow n(n-1)a^2 = 120$ Substitute $a = -\frac{12}{n}$: $n(n-1)\left(-\frac{12}{n}\right)^2 = 120$.

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43 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.

a. 2 ✓

b. 4

c. 3

d. 1

Step 1: Find the points of intersection of the curve and the line. Set the equations equal to each other: $6x - x^2 = mx \Rightarrow x^2 - (6 - m)x = 0 \Rightarrow x(x - (6 - m)) = 0$ Since the line intersects the curve at the origin $(0, 0)$ and at point P, the x-coordinate of P is $x = 6 - m$. Step 2: Set up the definite integral for the enclosed area. For $0 \leq x \leq 6 - m$, the curve lies above the straight line. The enclosed area A is given by.

44 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A particle moves in a straight line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 6t^2 - 18t + k$, where k is a constant. Given that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first 2 seconds.

a. 6 m ✓

b. 4 m

c. 5 m

d. 8 m

Step 1: Determine the unknown parameter k. The particle is momentarily at rest when $t = 2$, so $v(2) = 0$: $6(2)^2 - 18(2) + k = 0 \Rightarrow 24 - 36 + k = 0 \Rightarrow k = 12$ Step 2: Find all times when the particle is at rest to check for direction changes. $v(t) = 6t^2 - 18t + 12 = 6(t^2 - 3t + 2) = 6(t - 1)(t - 2) = 0$ The particle is at rest at $t = 1$ and $t = 2$. In the interval $0 \leq t0$; in the interval $1 < t < 2$, $v(t) < 0$. Step 3: Calculate the total distance traveled by integrating over each.

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45 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

The variables x and y satisfy the equation $y = Ab^x$, where A and b are positive constants. When $\log_2 y$ is plotted against x, a straight line is obtained that passes through the points $(2, 7)$ and $(5, 13)$. Find the value of y when $x = 4$.

a. **2048 ✓**

b. 1024

c. 11

d. 256

To reduce the non-linear relationship $y = Ab^x$ to linear form, take base-2 logarithms of both sides: $\log_2 y = \log_2 (Ab^x) = \log_2 A + x\log_2 b$ Setting $Y = \log_2 y$ and $X = x$, the equation represents a straight line $Y = mX + c$, where the gradient is $m = \log_2 b$ and the vertical intercept is $c = \log_2 A$. 1. Find the gradient m: $m = \frac{13-7}{5-2} = \frac{6}{3} = 2$ 2. Find the vertical intercept c using the point $(2, 7)$: $7 = 2(2) + c \Rightarrow c = 3$ Thus, the linear equation i.

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